

BUILD ARTIFACT 2: API CONTRACT APPENDIX

Endpoint definitions with request/response schemas | FastAPI | Decimal Technologies

WHY THIS EXISTS

The rule set specifies behavior (what each service decides) but not wire format (the JSON an engineer codes against). Per the architecture, business logic lives in FastAPI and n8n only routes events (ORCH-NOLOGIC-001). These contracts are the boundary between the two. Every endpoint maps to a module (M0-M19) and respects the Autonomy Ladder — endpoints that take external action return a recommendation for human approval at A1, not an executed action.

ENDPOINT INVENTORY

Endpoint	Module	Autonomy	Side Effect
POST /v1/signals	M1	A2	Stores signal, triggers scoring (no external action)
POST /v1/score	M2	A1	Computes scores, recommends tier (no auto-promote)
POST /v1/enrich	M3	A1	Runs enrichment pipeline for TIER_1 account
POST /v1/generate-message	M4	A1	Runs 7-agent chain, returns draft + Reasoning Passport
POST /v1/qc	M7	A1	Runs QC rubric, returns pass/fail + flagged claims
POST /v1/queue-message	M4	A1	Adds QC-passed msg to Ready-to-Fire Queue (QUEUE-ARCH-001)
POST /v1/send	M4	A1*	Executes send ONLY after human approval token present
POST /v1/engagement	M8	A2	Records open/click/reply, classifies intent
POST /v1/handoff	M9	A1	Builds handoff package, notifies sales
POST /v1/sync-crm	M11	A2	Bidirectional HubSpot sync per SOT rules
GET /v1/account/{id}	—	A2	Returns account + scores + state + decomposition
GET /v1/analytics/kpi	M10	A2	Returns KPI metrics (read-only)

SCHEMA: POST /v1/signals

REQUEST:

```
{
  "account_id": "uuid|null",          // null = entity resolution required
  "source": "google_news|sama_rss|linkedin|crunchbase|jobs|manual",
  "signal_type": "NEW_LICENSE|CTO_HIRING|FUNDING|OPEN_BANKING|...",
  "priority": "P1|P2|P3|P4",
  "raw_context": "string (full source text)",
  "source_url": "string",
  "detected_at": "ISO8601"
}
```

RESPONSE 201:

```
{
  "signal_id": "uuid",
  "is_duplicate": false,              // freshness>0.8 = duplicate (SIG-DEDUP)
  "confidence": 0.0-1.0,              // two-stage classification (FAIL-FALSE)
  "stored": true,
  "scoring_triggered": true,          // debounced 5-min window
  "trace_id": "uuid"                 // EVENT-RESP-004 lineage
}
```

SCHEMA: POST /v1/score

REQUEST: { "account_id": "uuid", "trigger_signal_id": "uuid" }

RESPONSE 200:

```
{
  "account_id": "uuid",
  "dynamic_score": 0-100,
  "ics": 0-100,
  "static_fit_score": 0-100,
  "dimension_breakdown": {           // EXPLAIN-002 decomposition
    "signal_strength": {"raw":75,"weight":0.35,"contribution":26.25},
    "regulatory":      {"raw":90,"weight":0.30,"contribution":27.0},
    "reachability":    {"raw":70,"weight":0.20,"contribution":14.0},
    "warmth":          {"raw":60,"weight":0.15,"contribution":9.0}
  },
  "effective_opportunity": float,    // full modifier chain applied
  "modifier_chain": {               // each value from Artifact 1
    "stage_bonus":1.3,"budget":0.5,"entrenchment":1.0,
    "risk":0.8,"window":1.5
  },
  "decision_score": float,
  "recommended_tier": "TIER_1|TIER_2|TIER_3",
  "tier_changed": true,
  "requires_human_approval": true   // A1: never auto-promotes
}
```

SCHEMA: POST /v1/generate-message

```
REQUEST: { "account_id":"uuid", "contact_id":"uuid", "signal_id":"uuid",
  "channel":"email|linkedin" }
RESPONSE 200:
{
  "message_id":"uuid", "draft":"string", "channel":"email",
  "reasoning_passport": {           // EXPLAIN-001 – required
    "signal":{...}, "interpretation":"...", "account_context":"...",
    "persona_analysis":"...", "pain_inference":"...",
    "strategy":"...", "qc_result":"..."
  },
  "evidence_strength": "FACT|INFERENCE|ASSUMPTION|SPECULATION",
  "verifiability_score": 0-100,      // AITRUST-003
  "qc_status":"PASSED|FAILED", "regeneration_count":0,
  "requires_human_approval": true
}
```

SCHEMA: POST /v1/send (the only externally-irreversible endpoint)

```
REQUEST:
{
  "message_id":"uuid",
  "human_approval_token":"uuid",    // REQUIRED. From reviewer action.
  "reviewer_id":"uuid",
  "send_window_check": true         // must pass KSA window (EXEC-006)
}
RESPONSE 200: { "sent":true, "sent_at":"ISO8601",
  "accountability_record_id":"uuid" } // HUMANREVIEW-001
ERROR 403: { "sent":false,
  "reason":"NO_APPROVAL_TOKEN|OUTSIDE_SEND_WINDOW|",
  "CONTACT_EXHAUSTED|RAMADAN_BLACKOUT" }
```

CONTRACT RULES

1. Every endpoint that could take external action returns `requires_human_approval` at A1. Only `/v1/send` executes, and only with a valid `human_approval_token`.
2. Every response carries a `trace_id` for end-to-end lineage (EVENT-RESP-004).
3. Scores always return their decomposition — no black boxes (EXPLAIN-002).
4. All endpoints are idempotent on (`trace_id`, `operation`) — replaying an event produces the same state (Meta-Principle event-driven idempotency).
5. Errors are structured with a machine-readable reason enum, never a bare 500.